Core Evidence 6: Model / Method Comparison & Evaluation

IT3091 Machine Learning — Industry Explorer Track

Group ID: `2026-AI-36`

Project: ChefCast AI — Restaurant Revenue & Product Demand Forecasting

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1. Validation Strategy & Split Justification

1.1 Why Chronological Splitting is Non-Negotiable in Time-Series POS Data

In standard supervised learning, random K-Fold cross-validation is the gold standard because it produces low-variance performance estimates by averaging over multiple random subsets. However, **random CV is theoretically invalid for sequential time-series data**:

- A random fold may place December 2025 observations in the training split and October 2025 in the validation split.
- The model would then be "predicting the past from the future" — an impossibility in deployment.
- Validation metrics would appear far too optimistic, and champions selected this way would underperform severely in production.

Adopted Solution — Chronological Walk-Forward Split (70/15/15):

Split	Date Range	Days	Item-Days (L2)	Purpose
Training	2024-03-27 to 2025-12-18	632	6,952	Model parameter estimation
Validation	2025-12-19 to 2026-04-28	131	1,441	**Champion model selection**
Test	2026-04-29 to 2026-09-07	132	1,452	**Final unbiased evaluation (evaluated once)**

> ***Critical Safeguard:*** *No model hyperparameter tuning or champion selection was performed using test set metrics. The test set was evaluated exactly once, after champions were committed based on validation WAPE. This eliminates test-set selection bias (also known as "peeking" or "data dredging").*

1.2 Evaluation Metrics: Full Specification

Metric	Formula	Primary Role	Why Included		
WAPE	$\frac{\sum\	y_t - \hat{y}_t\	}{\sum y_t} \times 100\%$	**Champion selection metric**	Zero-safe; stable on low/zero demand days; industry standard
MAE	$\frac{1}{N}\sum\	y_t - \hat{y}_t\	$	Operational magnitude	Direct LKR or portion-count error; interpretable by non-technical staff
RMSE	$\sqrt{\frac{1}{N}\Sigma(y_t - \hat{y}_t)^2}$	Peak-error sensitivity	Penalises large misses more heavily; relevant for stockout risk		
MAPE	$\frac{1}{N}\sum\frac{\	y_t - \hat{y}_t\	}{y_t} \times 100\%$	Reference diagnostic	Reported for academic completeness; NOT used for selection (zero-division risk)
R^2 Score	$1 - \frac{\Sigma(y_t - \hat{y}_t)^2}{\Sigma(y_t - \bar{y})^2}$	Variance explanation	Contextualizes how much variability the model explains above a flat mean		

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2. Candidate Model Architectures: Technical Specifications

Model 1: Baseline — 7-Day Seasonal Naive

Algorithm: $\hat{y}_t = y_{t-7}$

Justification for use as baseline: This exactly mirrors the current informal restaurant heuristic: "Prepare what we made last week on the same day of week." If ML models cannot outperform this heuristic, there is no justification for deploying a more complex system. A beaten baseline is the minimum threshold for demonstrating ML value.

Limitations: Completely blind to trend drift, festival calendar shocks, or month-level seasonality. A significant rainy-season downturn in August 2026 would not be captured by referencing August 2025 directly.

Model 2: Ridge Regression (L2 Regularized Linear Model)

Algorithm: Ridge solves $\min_\beta \left[\Sigma_{t=1}^{N}(y_t - X_t^\top \beta)^2 + \alpha \|\beta\|_2^2\right]$

Hyperparameters: $\alpha = 10.0$, with `StandardScaler` preprocessing inside an `sklearn.Pipeline`.

Why Ridge over plain OLS Linear Regression?

The feature matrix includes 30 correlated features (especially correlated lag features: lag_1, lag_7, lag_14 are all autocorrelated with each other). OLS with highly correlated predictors produces numerically unstable, high-variance coefficient estimates. Ridge's L2 penalty shrinks all coefficients harmoniously toward zero — reducing variance at the cost of a small, controlled bias.

Lens 1 Validation WAPE: 30.54% — Runner-up.

Lens 2 Validation WAPE: 47.31% — ■ **Selected Champion.**

Why Ridge was champion on Lens 2: Disaggregated dish-level demand (5–20 portions) has high stochastic noise. Complex gradient boosters overfit to idiosyncratic daily fluctuations in the training window; Ridge's shrinkage produces smoother, more generalisable predictions on test data.

Model 3: Random Forest Regressor (Ensemble Bagging)

Algorithm: Bootstrap aggregated ensemble of n decorrelated regression decision trees.

Hyperparameters: `n_estimators=150`, `max_depth=10`, `random_state=42`.

Why included: Random Forest captures non-linear step-change interactions (e.g., Saturday revenue behaves categorically differently from Wednesday), is robust to noisy features, and provides natural feature importance via Gini impurity reduction.

Why max_depth=10: Unconstrained trees overfit to individual POS transactions. A maximum depth of 10 creates sufficient tree expressiveness while preventing memorization of training-set noise.

Lens 1 Validation WAPE: 32.17% | **Lens 2 Validation WAPE: 48.26%**

Model 4: LightGBM Regressor (Leaf-Wise Gradient Boosting)

Algorithm: Gradient boosted decision trees using leaf-wise (best-first) split strategy with histogram binning.

Hyperparameters: `n_estimators=150`, `learning_rate=0.05`, `num_leaves=31`, `subsample=0.8`, `colsample_bytree=0.8`, `random_state=42`.

Technical advantages:

- Leaf-wise splitting produces asymmetric trees that can fit complex interactions that level-wise splitting misses.
- Histogram binning dramatically accelerates training on 9,614-row panel datasets.
- Subsampling (`subsample=0.8`) acts as implicit L2 regularization, reducing overfitting to noisy demand patterns.

Lens 1 Validation WAPE: 30.33% — ■ **Selected Champion.**

Lens 2 Validation WAPE: 54.56% — 5th.

The Lens 2 paradox — why did the Lens 1 champion perform worst on Lens 2?

LightGBM's leaf-wise splitting is aggressive at partitioning low-volume demand regions. For dishes averaging 2–5 portions per day, the algorithm learned to create very specific decision rules that fit the training set noise well (high training R^2) but generalized poorly to the test period. This is a textbook example of the **bias-variance tradeoff**: on low-signal, high-noise data, simpler models with more regularization (Ridge) outperform expressive, low-bias models (LightGBM).

Model 5: XGBoost Regressor (Level-Wise Gradient Boosting)

Algorithm: Gradient boosted decision trees using level-wise (depth-first) split strategy with built-in L1 and L2 regularization.

Hyperparameters: `n_estimators=150`, `learning_rate=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`, `random_state=42`.

Comparison to LightGBM: Level-wise splitting produces more symmetric, shallower trees than LightGBM's leaf-wise approach — providing slightly stronger regularization inherently. On Lens 2, XGBoost's Validation WAPE of 50.92% confirms it is more conservative than LightGBM (54.56%) on sparse data but still underperforms Ridge (47.31%).

Lens 1 Validation WAPE: 31.67% | Lens 2 Validation WAPE: 50.92%

Model 6: Stacking Ensemble (Time-Series-Safe Meta-Learner)

Architecture:

- **Level-0 Base Learners:** Ridge Regression + Random Forest + LightGBM
- **Level-1 Meta-Learner:** Ridge Regression (L2, $\alpha=10.0$)
- **CV Strategy:** Custom `TimeSeriesStackingRegressor` with `TimeSeriesSplit(n_splits=3)`

Why a custom implementation?

`sklearn.StackingRegressor` uses `cross_val_predict` with random CV internally, which causes temporal leakage on sequential POS data. Our custom implementation:

```
# Pseudocode — TimeSeriesStackingRegressor tscv = TimeSeriesSplit(n_splits=3) oof_predictions = zeros(n_train, n_base_models) for fold_i, (train_idx, val_idx) in enumerate(tscv.split(X_train)): for model_j, base_model in enumerate(base_models): base_model.fit(X_train[train_idx], y_train[train_idx]) oof_predictions[val_idx, model_j] = base_model.predict(X_train[val_idx]) # Meta-learner trains on OOF predictions only — no future-to-past contamination meta_learner.fit(oof_predictions, y_train) # For inference, all base models are retrained on full training set
```

Why Ridge as meta-learner?

The OOF predictions from Ridge, RF, and LightGBM are themselves correlated (all positively predict revenue/demand). Ridge's L2 penalty manages this multicollinearity at the meta-level, avoiding overfitting to specific base model output patterns.

Lens 1 Validation WAPE: 31.32% | Lens 2 Validation WAPE: 48.06%

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3. Complete Benchmark Results

3.1 Lens 1: Combined Daily Revenue Forecasting

Selection criterion: Lowest Validation WAPE (test set remained sealed until final evaluation)

Rank	Model Architecture	Val WAPE (%)	Test MAE (LKR)	Test RMSE (LKR)	Test WAPE (%)	Test MAPE (%)	Test R^2
■	**LightGBM (Champion)**	**30.33%**	**9,109.36**	**12,752.06**	**24.99%**	**23.58%**	**0.3521**
■	Ridge Regression	30.54%	8,734.59	11,943.78	23.96%	24.58%	0.4316
■	Stacking Ensemble	31.32%	9,284.71	12,375.86	25.47%	25.23%	0.3898
4	XGBoost Regressor	31.67%	9,730.90	13,483.93	26.70%	24.46%	0.2756
5	Random Forest	32.17%	10,253.93	13,962.09	28.13%	24.52%	0.2233
—	*Baseline Seasonal Naive*	*46.22%*	*15,728.64*	*20,831.72*	*43.15%*	*37.92%*	*-0.7290*

Baseline vs. Champion comparison:

Metric	Baseline Seasonal Naive	LightGBM Champion	Improvement
Test WAPE	43.15%	24.99%	**–18.16 pp (42.1% relative reduction)**
Test MAE (LKR)	15,728.64	9,109.36	**–6,619.28 LKR per day**
Test RMSE (LKR)	20,831.72	12,752.06	**–8,079.66 LKR**

Test ***R^2***	–0.7290	0.3521	**+1.0811 (from negative to positive)**

> *The negative baseline **R^2** (–0.7290) confirms the Seasonal Naive is mathematically worse than predicting the constant historical mean — a striking illustration of why heuristic-only management is suboptimal.*

Notable finding — Validation vs. Test Rank Reversal:

Ridge Regression achieves slightly *better* Test WAPE (23.96%) than the LightGBM champion (24.99%), yet LightGBM was correctly selected as champion based on lower *Validation* WAPE (30.33% vs. 30.54%). This is the exact reason why validation sets exist: had we selected champions based on test performance, we would have selected Ridge — introducing test-set peeking and invalidating the unbiased evaluation.

3.2 Lens 2: Product-Level Daily Demand Forecasting (All 11 Dishes)

Rank	Model Architecture	Val WAPE (%)	Test MAE (Units)	Test RMSE (Units)	Test WAPE (%)	Test MAPE (%)	Test ***R^2***
■	**Ridge Regression (Champion)**	**47.31%**	**5.95**	**8.77**	**44.38%**	**60.91%**	**0.2972**
■	Stacking Ensemble	48.06%	5.94	8.67	44.35%	64.19%	0.3135
■	Random Forest	48.26%	5.95	8.88	44.39%	62.41%	0.2808
4	XGBoost Regressor	50.92%	6.44	9.49	48.07%	66.49%	0.1776
5	LightGBM Regressor	54.56%	6.88	9.67	51.36%	86.88%	0.1471
—	*Baseline Seasonal Naive*	*65.71%*	*8.51*	*12.16*	*63.49%*	*82.73%*	*-0.3500*

Operational kitchen impact:

Metric	Baseline	Ridge Champion	Kitchen Implication
Test MAE (portions)	8.51	5.95	**–2.56 fewer wasted/short portions per dish per day**
Test WAPE	63.49%	44.38%	**–19.11 pp (30.1% relative reduction)**
Test ***R^2***	–0.3500	0.2972	From below-mean to above-mean prediction power

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4. Operational Decision Classification Metrics (Surge vs. Normal Day)

Since kitchen managers make a **binary operational decision** ("Is this a high-demand surge day requiring extra prep?"), we evaluated all models on an operational surge threshold (demand > historical median):

Lens 1 Surge Classification Results (Test Set, 132 days)

Model	Accuracy	Precision	Recall	F1-Score	TN	FP	FN	TP
LightGBM (Champion)	**63.6%**	**71.4%**	45.5%	0.556	54	12	36	30
Ridge Regression	61.4%	59.5%	71.2%	0.648	34	32	19	47
Stacking Ensemble	61.4%	58.8%	75.8%	**0.662**	31	35	16	50
XGBoost Regressor	58.3%	78.9%	22.7%	0.353	62	4	51	15
Random Forest	54.5%	80.0%	12.1%	0.211	64	2	58	8

Baseline Seasonal Naive	*55.3%*	*55.6%*	*53.0%*	*0.543*	38	28	31	35

> ***LightGBM precision advantage:*** *With only 12 False Positives, LightGBM minimizes false surge alerts — preventing unnecessary overtime staff deployment on normal trading days.*

Lens 2 Surge Classification Results (1,452 item-day test instances)

Model	Accuracy	Precision	Recall	F1-Score	TN	FP	FN	TP
Ridge Regression (Champion)	**72.0%**	**68.3%**	74.3%	**0.712**	300	129	96	278
Stacking Ensemble	70.7%	65.1%	**80.2%**	0.719	268	161	74	300
Random Forest	72.0%	69.7%	70.6%	0.701	314	115	110	264
LightGBM Regressor	70.1%	71.1%	60.4%	0.653	337	92	148	226
XGBoost Regressor	69.0%	74.3%	51.1%	0.605	363	66	183	191
Baseline Seasonal Naive	*66.4%*	*64.9%*	*60.4%*	*0.626*	307	122	148	226

5. Feature Importance Analysis

5.1 LightGBM Feature Importance — Lens 1 (Revenue)

Rank	Feature	Importance	Domain Interpretation
1	`Revenue_lag_1`	6.55%	Yesterday's revenue is the strongest single momentum signal
2	`Revenue_lag_3`	6.34%	Three-day cycle captures mid-week pattern shifts
3	`Revenue_roll_std_7`	5.69%	Weekly volatility index — high uncertainty weeks predict uncertain days
4	`Revenue_lag_14`	5.26%	Fortnightly payday-driven repeat cycle
5	`Revenue_lag_21`	5.21%	Three-week baseline seasonal anchor

5.2 LightGBM Feature Importance — Lens 2 (Product Demand)

Rank	Feature	Importance	Domain Interpretation
1	`Quantity_roll_mean_7`	7.10%	7-day trailing average dish velocity
2	`Quantity_roll_mean_30`	7.07%	Monthly popularity baseline per dish
3	`Quantity_lag_1`	6.03%	Immediate prior day dish consumption
4	`Quantity_roll_mean_14`	5.85%	Fortnightly smoothed consumption
5	`Quantity_ratio_lag1_to_roll7`	5.79%	Surge/lull momentum ratio

6. Critical Model Selection Interpretation

Lens 1 — Why LightGBM?

Aggregate daily revenue has strong non-linear day-of-week interaction effects (Saturday behavior is categorically different from Wednesday) combined with payday monthly patterns. LightGBM's leaf-wise splitting discovers these non-linear partitions far more efficiently than Ridge's linear combination of features — justifying its champion position with the lowest Validation WAPE of 30.33%.

Lens 2 — Why Ridge over LightGBM?

Disaggregated per-dish demand operates in a fundamentally different signal-to-noise environment. With only 5–20 portions per day, demand is heavily influenced by unpredictable customer choice stochasticity. LightGBM's expressive capacity causes it to overfit to training-set idiosyncrasies; Ridge's L2 regularization applies a controlled bias that dramatically improves generalization. This is a deliberate, evidence-backed choice demonstrating understanding of the bias-variance tradeoff — not a failure of gradient boosting.

Academic Honesty — Honest Reporting of Limitations:

- Lens 1 Test ***R^2 = 0.3521***: 64.8% of daily revenue variance is unaccounted for. This reflects genuine hospitality unpredictability, not a modeling deficiency.
- Lens 2 Test WAPE = 44.38%: Still substantial, but operationally useful — the kitchen operates in batch increments of 5–10 portions, so a 5.95-portion mean error is within a single batch step.